

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Butyrylcholinesterase Activity & Concentration (BCHE_L)

Data File: BCHE_L.xpt

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Component Description

Butyrylcholinesterase (BChE) is an enzyme primarily synthesized in the liver and found in high concentrations in blood plasma. BChE can be used as an indicator of reduced cholinesterase (ChE). Due to its higher concentrations in blood plasma, because it is freely circulating in blood, and can be obtained directly from either serum or plasma, BChE activity is often a better indicator for acute ChE inhibition (Bartels et. al., 2000). While BChE is naturally produced, its activity can be affected by environmental exposures. BChE activity measurements can be used to indicate various health and exposure conditions including liver dysfunction, drug sensitivity, and exposure to organophosphate (OP) pesticides and nerve agents (Perez et. al. 2015).

NHANES offers a unique opportunity to determine the baseline for blood BChE biomarkers (butyrylcholinesterase activity and butyrylcholinesterase concentration) that will serve as a clinically significant indicator of cholinesterase inhibition and establish references ranges for the general population.

Eligible Sample

Examined participants aged 6 years and older were eligible.

Description of Laboratory Methodology

Butyrylcholinesterase Activity

The method described here quantifies BChE activity in human serum/plasma using absorption spectroscopy to monitor the rate of substrate hydrolysis (Ellman et. al., 1961) and (Munro et. al., 1994). Samples are incubated with butyrylthiocholine (BTCh) and dithiobisnitrobenzoic acid (DTNB). During this incubation, active BChE will hydrolyze BTCh into butyrate and thiocholine. The thiol generated by the formation of thiocholine rapidly reacts with DTNB cleaving the disulfide bond to form nitrothiobenzoate (NTB) which presents an absorption band centered at 412 nm. Monitoring the absorption at 412 nm during the incubation provides information on the rate at which NTB is generated which occurs at a 1:1 ratio with the formation of thiocholine.

Butyrylcholinesterase Concentration

The method described here utilizes immunomagnetic separation liquid chromatography-tandem mass spectrometry (LC-MS/MS) to quantify unadducted BChE (Knaack et. al., 2012). Unadducted BChE measurement can quickly identify samples with reduced levels of the endogenous protein. Immunomagnetic beads conjugated to monoclonal BChE antibodies specifically purify both unadducted and OP-adducted BChE protein from serum/plasma. BChE protein bound to the bead is then digested with pepsin to yield a nine amino acid biomarker peptide containing the BChE active site. Unadducted BChE has a unique chemical structure

that can be chromatographically separated from adducted BChE and detected by tandem mass spectrometry.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

This is a new component for the NHANES August 2021-August 2023 cycle.

Laboratory Method Files

[Butyrylcholinesterase Activity Laboratory Procedure Manual](#) (August 2026)

[Butyrylcholinesterase Concentration Laboratory Procedure Manual](#) (August 2026)

Laboratory Quality Assurance and Monitoring

Serum specimens were processed, stored, and shipped to Division of Laboratory Sciences, National Center for Environmental Health, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate refrigerated (2-8°C) conditions until they are shipped to National Center for Environmental Health for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Amendment mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the QC procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et. al., 1981) when testing NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

All QC procedures recommended by the manufacturers were followed. Reported results for all assays meet the Division of Laboratory Sciences' QA/QC performance criteria for accuracy and precision, similar to the Westgard rules (Caudill et. al., 2008).

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed.

Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for details on the use of sample weights and analytic issues.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES August 2021-August 2023 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

The [Fasting Questionnaire File](#) includes auxiliary information such as fasting status, length of fast and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection limits

The lowest reportable limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending in "LC" (ex., LBXBALC) indicates whether the result was below the lowest reportable limit: "0" means that the result was at or above the lowest reportable limit, "1" indicates that the result was below the lowest reportable limit. The other variable prefixed LBX (ex., LBXBA) provides the analytic result for that analyte. For analytes with analytic results below the lowest reportable limit (ex., LBXBALC=1), an imputed fill value was placed in the analyte results field. This value is the lowest reportable limit divided by square root of 2 (LRL/sqrt [2]).

Lowest Reportable Limit (LRL, in ng/mL) for Serum:

VARIABLE NAME	SAS LABEL	LRL
LBXBA	Butyrylcholinesterase Activity (U/mL)	0.05
LBXCH	Butyrylcholinesterase Con (ng/mL)	2

References

- Barterls, C.F.; Xie, W.; Miller-Lindholm, A.K.; Schoper, L.; Lockridge, O.; Biochem Pharmacol, 2000, 60, 479-487
- Caudill, S.P., Schleicher, R.L., Pirkle, J.L. Multi-rule quality control for the age-related eye disease study. Statist. Med. (2008) 27(20):4094-40106.
- Ellman, G. L., Courtney, K. D., Andres Jr, V., & Featherstone, R. M. (1961). A new and rapid colorimetric determination of acetylcholinesterase activity. Biochemical Pharmacology, 7(2), 88-95. DOI: 10.1016/0006-2952(61)90145-9. Worek, F.; Mast, U.; Kiderlen, D.; Diepold, C.; Eyer, P.

(1999). Improved determination of acetylcholinesterase activity in human whole blood. *Clinica Chimica Acta*, 288, 73-90. DOI: 10.1016/s0009-8981(99)00144-8

- Knaack, J.S. (2012). A high-throughput diagnostic method for measuring human exposure to organophosphorus nerve agents. *Anal Chem*, 84(21), 9470-7. DOI: 10.1021/ac302301w.
- Munro, N. (1994). Toxicity of the organophosphate chemical warfare agents GA, GB, and VX: implications for public protection. *Environmental Health Perspectives*, 102(1), 18-37. DOI: 10.1289/ehp.9410218
- Perez, J.W.; Pantazides, B.G.; Watson, C.M.; Thomas, J.D.; Blake, T.A.; Johnson, R.C.; *Analytical Chemistry*, 2015, 87, 5723-5729. DOI: 10.1021/acs.analchem.5b00893
- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. *Clin Chem* (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

LBXBA - Butyrylcholinesterase Activity (U/mL)

Variable Name: LBXBA

SAS Label: Butyrylcholinesterase Activity (U/mL)

English Text: Butyrylcholinesterase Activity (U/mL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1.02 to 8.99	Range of Values	7061	7061	
.	Missing	1007	8068	

LBDBALC - Butyrylcholinesterase Act comt code

Variable Name: LBDBALC

SAS Label: Butyrylcholinesterase Act comt code

English Text: Butyrylcholinesterase Activity (U/mL) Comment Code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7061	7061	
1	Below lower detection limit	0	7061	
.	Missing	1007	8068	

LBXCH - Butyrylcholinesterase Con (ng/mL)

Variable Name: LBXCH

SAS Label: Butyrylcholinesterase Con (ng/mL)

English Text: Butyrylcholinesterase Concentration (ng/mL)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
15.73 to 194.14	Range of Values	7060	7060	
.	Missing	1008	8068	

LBDCHLC - Butyrylcholinesterase Conc comt code

Variable Name: LBDCHLC

SAS Label: Butyrylcholinesterase Conc comt code

English Text: Butyrylcholinesterase Concentration (ng/mL) Comment Code

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
0	At or above detection limit	7060	7060	
1	Below lower detection limit	0	7060	
.	Missing	1008	8068	